

Dual-Array Passive Acoustic Mapping for Cavitation Imaging With Enhanced 2-D Resolution

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Abstract—Passive acoustic mapping (PAM) techniques have been developed for the purposes of detecting, localizing, and quantifying cavitation activity during therapeutic ultrasound procedures. Implementation with conventional diagnostic ultrasound arrays has allowed planar mapping of bubble acoustic emissions to be overlaid with B-mode anatomical images, with a variety of beamforming approaches providing enhanced resolution at the cost of extended computation times. However, no passive signal processing techniques implemented to date have overcome the fundamental physical limitation of the conventional diagnostic array aperture that results in point spread functions with axial/lateral beamwidth ratios of nearly an order of magnitude. To mitigate this problem, the use of a pair of orthogonally oriented diagnostic arrays was recently proposed, with potential benefits arising from the substantially expanded range of observation angles. This article presents experiments and simulations intended to demonstrate the performance and limitations of the dual-array system concept. The key finding of this study is that source pair resolution of better than 1 mm is now possible in both dimensions of the imaging plane using a pair of 7.5-MHz center frequency conventional arrays at a distance of 7.6 cm. With an eye toward accelerating computations for real-time applications, channel count reductions of up to a factor of eight induce negligible performance losses. Modest sensitivities to sound speed and relative array position uncertainties were identified, but if these can be kept on the order of 1% and 1 mm, respectively, then the proposed methods offer the potential for a step improvement in cavitation monitoring capability.

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I. INTRODUCTION

AS THE scope of therapeutic ultrasound applications in clinical medicine continues to expand in areas ranging from pain reduction [1], [2] to drug delivery to the brain [3] and solid tumors [4], [5], so too does the need for treatment monitoring methods that can help reduce risk and enhance efficacy. Current noninvasive options include techniques based on magnetic resonance (thermometry, elastography) and ultrasound (conventional B-mode, elastography, passive cavitation detection) [6]. For treatments where acoustic cavitation is to be maximized, moderated, or avoided, passive ultrasound schemes offer a relatively simple means for detection, localization, and quantification of cavitation activity. Absent the transmissions required for active methods, passive techniques may provide a cavitation monitoring capability at any time during the application of a therapeutic ultrasound field, without concern for waveform interference [7].

Early studies of passive cavitation detection employed a single element receiver that enabled identification of characteristic bubble emissions within the detector beam pattern [8], [9]. Spatial selectivity was improved using multiple detectors with overlapping beams [10]. More flexible approaches employed conventional diagnostic arrays [11], [12], allowing the temporal, spatial, and spectral characteristics of individual cavitation events to be separated and tracked over the course of a therapeutic procedure. By implementing array-based processing schemes, passive acoustic mapping (PAM) allows emitted acoustic energy in a specified frequency band to be quantified and overlaid with anatomic information from conventional B-mode or other imaging modalities.

PAM spatial resolution has been further enhanced using data-dependent adaptive routines [13], [14] and other improved beamformers [15], [16]. The advent of hemispherical arrays has provided high-resolution 3-D cavitation imaging [17], [18], but with a substantial increase in cost and decrease in portability. In our laboratory, cost and resolution tradeoff considerations have favored the use of a conventional handheld imaging probe and data-adaptive beamforming for imaging tasks that do not involve propagation through bone.

Still, no matter the beamformer employed, PAM implemented with a conventional linear or curvilinear array will suffer from axial resolution that may be an order of magnitude

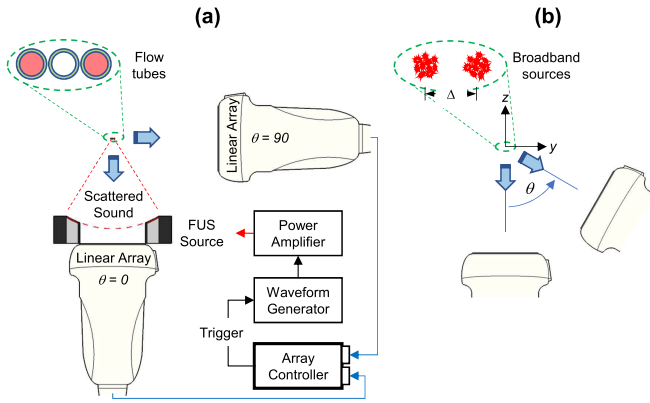


Fig. 1. (a) Experiment and (b) simulation configurations. In (a), the red tube fill color indicates contrast agent, and in (b), red symbols indicate simulated sources. Although two FUS devices were deployed, only one (shown) was used to initiate cavitation. In the experiments, contrast agent flow was normal to the page, and in the simulations, the length axis of each cylindrical bounding source region was oriented normal to the imaging plane. Unless otherwise stated, the arrays were located 7.6 cm from the center of the group of tubes.

coarser than the lateral resolution—the ratio of which varies linearly with increasing depth [19]. For example, the -3 -dB transverse and axial beamwidths of a 4-cm-wide linear array (e.g., L11-4) imaging a 7-MHz point source at 7 cm depth with a conventional beamformer are calculated to be 0.34 and 4.7 mm, respectively. In the frequency range where conventional arrays are both available and broadly applicable to clinical work (<10 MHz for depths of several cm), the ability to localize and separate cavitation events may not be adequate either for laboratory or clinical application. In particular, submillimeter feature localization would be desirable for detailed studies of intra- and extra-vascular bubble behavior and particle transport [20], [21], particularly in murine preclinical models where the tumor diameter is commonly 4–10 mm [22].

In the continuing quest to improve the resolution of PAM techniques while staying within existing financial and computational budgets, beamforming with an orthogonally oriented pair of co-planar diagnostic arrays has recently been explored [23] (Fig. 1). It was hypothesized that the expanded aperture of the dual-array configuration would provide the desired submillimeter resolution in both coordinates of the imaging plane using existing hardware and imaging routines. This article presents evaluations of the dual-array PAM concept carried out through simulations and experiments, with an emphasis on source resolvability in two dimensions. The study includes assessments of sensitivity to uncertainties in array positioning and sound speed, and a method for accelerating image computation times is also investigated with the aim of improving image update rates for real-time monitoring.

II. METHODS

A. Experiments

In order to determine the impact of the added array on PAM resolution, a series of experiments was conducted with a pair of identical diagnostic arrays to monitor cavitation emanating from one or more tubes containing a flowing ultrasound cavitation agent. Experiments were conducted

using the configuration shown in Fig. 1(a), with all transducers fully immersed in a tank filled with filtered and degassed water at temperatures between 19 °C and 21 °C.

1) *Arrays and Data Collection*: This study employed a pair of L11-4v arrays (Verasonics, Kirkland, WA, USA), each having 128 elements with 0.30-mm spacing and an operating band of 4–11 MHz. Array data sets were collected using a 256-channel configurable ultrasound array controller (Vantage 256, Verasonics), which was also used to provide synchronization pulses to auxiliary data collection equipment.

2) *Test Configuration*: A single-element focused ultrasound (FUS) transducer was employed as a source for cavitation generation. The transducer (H107, Sonic Concepts, Woodinville, WA) had a fundamental frequency of 0.5 MHz, 64-mm active diameter, 62.6-mm spherical geometric focus, and full-width at half-maximum (FWHM) beamwidths of 3.2 (lateral) and 28.4 mm (axial). The FUS transducer also had a 47×20 mm² rectangular opening to facilitate mounting of and co-aligned imaging with a single L11-4v array.

Before testing, a thin layer of ultrasound gel (Anagel, Ana Wiz Ltd., Surrey, U.K.) was applied over the array lenses, and the devices were fully enclosed in a latex cover for waterproofing (Ecolab Ultracover, Microtek Medical, Zutphen, The Netherlands).

The array pair was configured with a 90° separation angle in the imaging (yz) plane by mounting each array to the back of an FUS transducer, each of which was held in a Delrin thermoplastic “L” bracket (not shown) that set the device orientations and radii to a common focal point [21]. This orthogonal arrangement was expected to provide superior image quality by applying the fine inherent lateral resolution of each array to both axes of the imaging plane. Although two FUS devices were deployed for this configuration, only one was used to initiate cavitation. The devices were aligned using a needle hydrophone to locate the FUS focus, and subsequently by adjusting the array positions so that the hydrophone tip was centered within B-mode images for each array.

Once the transducers were aligned, sets of three equal diameter, equally spaced polyethylene tubes (SIMS Portex Ltd., Hythe, U.K.) were collinearly installed parallel to the (y) imaging axis so that flow was normal to the imaging plane, and the middle tube was centered in each of the individual-array B-mode images. Tubing was held in a Perspex fixture (not shown) such that holder scattering would be minimized. The tube diameters (0.3/0.6, 0.4/0.8, 0.5/1.0 mm, inner/outer) were varied in order to evaluate limits of cavitation region resolvability.

Tubes were filled either with degassed water or an ultrasound contrast agent (SonoVue, Bracco, Milan, Italy), the latter being used to provide a renewable population of microbubbles. Tubes containing undiluted SonoVue were supplied by 1-mL syringes and operated manually during each experiment. Before each experiment, each syringe was refilled and lightly shaken in order to remix the suspension. The use of undiluted SonoVue was intended to maximize cavitation signal amplitudes observed when using the smallest tubes and thereby enable analysis of small tube pair resolvability.

TABLE I
SIMULATION PARAMETERS

Parameter	Value	Justification
Cavitation Sources		
Sources per Tube	20	Reproduces experiment data trends
Bounding Diameter	0.3 – 0.5 mm	Matches tubes in experiments
Bounding Height	3.4 mm	Matches FUS field FWHM
Cavitation Signals		
Amplitude	1.0 ± 0.5 Pa	Reproduces experiment data trends
Delay	0.0 ± 0.10 μs	Reproduces experiment data trends
FUS Excitation		
Frequency	500 kHz	Matches FUS used in experiments
Pulse length	100 μs	Reproduces experiment data trends with reduced computation time

Note: values written as $xx \pm yy$ express mean and standard deviation for randomized quantities. Otherwise, a fixed value was used.

Cavitation measurements were performed by driving the FUS transducer with a 200-cycle tone burst provided by a waveform generator (model 33250A, Agilent Technologies, Cheshire, U.K.) with a 10-Hz pulse repetition frequency, passed on to a power amplifier (model A300, Electronics and Innovation Ltd., Rochester, NY, USA), and on to the FUS matching transformer and the transducer itself. The choices of FUS signal duration and repetition rate were made in an effort to balance the tradeoff between signal energy per pulse and efficient use of the microbubble suspension. The FUS drive level was chosen to produce a peak negative focal pressure of approximately 2.0 MPa, which reliably produced broadband emissions from the SonoVue-filled tubes. Received signals were synchronously acquired from both arrays with a sample rate of 25 MHz and stored as MATLAB files (Mathworks, Natick, MA, USA).

B. Array Simulations

Companion simulations were run both to establish a model for dual-array beamforming and to investigate parameter sensitivities such as array angular separation and sound speed uncertainty. The simulation geometry is shown in Fig. 1(b). Time domain cavitation-like signals were generated using a simplified version of the Vokurka model [13], [24] that employed a reduced set of input variables (Appendix A). A set of point sources was randomly distributed within a cylindrical region approximating the interior volume of each tube employed in the experiments. The length axis of each bounding cylinder was oriented orthogonally to the PAM imaging plane, mimicking the direction of contrast agent flow in the experiments. The height of the bounding cylinder was defined to match the FWHM of the FUS field. Simulation parameter values are listed in Table I, and many were chosen to generate signal characteristics that qualitatively mimic trends observed in the data (see Appendix A).

Simulated bubble emission signals in the form of randomly scaled and delayed impulses were filtered by the typical array element frequency response [25] and propagated from each source to each array element assuming spherical spreading. The total element response was found by summing the contributions from all sources. Finally, a spectrally white noise floor was added to yield array element signal-to-noise ratio

(SNR) values representative of those seen in the experiments—typically 15 dB. A minimum of ten realizations of the randomized variables above were run for calculation of performance statistics. Examples of simulated and measured signals from a single tube experiment are shown in Appendix A.

C. PAM Algorithms

Cavitation emissions received with conventional ultrasound linear arrays were processed with two PAM algorithms that have been adopted for cavitation monitoring: time exposure acoustics (TEA) [7] and robust Capon beamforming (RCB) [13]. The former provides a computationally simple method of mapping sources of cavitation at the expense of spatial resolution, while the latter provides superior resolution but at a higher computational cost.

Both algorithms operate on time domain array element response signals $p(t, \mathbf{r}_j)$ received at array element locations $\mathbf{r}_j$ for time duration T to produce an estimate of monopolar source energy $E(\mathbf{r})$ at an observation location $\mathbf{r}$

$$E(\mathbf{r}) = \frac{1}{4\pi\rho c} \int_{t_0}^{t_0+T} q^2(\mathbf{r}, t) dt \quad (1)$$

where ρ and c are the density and sound speed of the ambient medium, respectively, t_0 is the data record start time, and bold type quantities indicate vectors. The source strength q is defined by a sum over the $j = 1 : J$ array elements

$$q(\mathbf{r}, t) = \frac{1}{J} \sum_{j=1}^J 4\pi w_j |\mathbf{r}_s| p(\mathbf{r}_j, t + |\mathbf{r}_s|/c) \quad (2)$$

where $\mathbf{r}_s = \mathbf{r}_j - \mathbf{r}$ is the distance from the j th element to the observation point, $|\mathbf{r}_s|/c$ is a geometric steering delay, and w_j are the weights applied to each element.

All weights in the TEA beamformer are set to unity, while RCB optimizes the weights for each imaging location in order to minimize the variance of the array output, with allowance for uncertainties in array element sensitivity and location. Control over the degree to which weights may be adjusted is provided by a single input factor ϵ . The larger the value of ϵ (a real, positive number), the greater the flexibility allowed in array weight adjustment. A value of $\epsilon = 1$ was used for simulations of individual and paired arrays, which were conducted assuming ideal intact arrays. A value of 24 was used for the experiments after a preliminary processing review of the data, and this higher value is consistent with prior studies [26] conducted with uncalibrated arrays. A broader discussion of epsilon, its selection, and its impact on algorithm performance can be found in [13] and [27].

In order to minimize the influence of FUS harmonic scattering and reduce processing noise, all time domain signals were processed with a fifth-order 4.0–12.0-MHz Butterworth band-pass filter prior to beamforming. Measured signals from array elements that were nonresponsive (exhibiting only electronic noise) were not passed to the beamformers.

Data handling for single- and dual-array imaging was identical: the beamformer received the time domain data from all responsive array elements, their locations, the image domain (span and resolution), and an assumed global

sound speed. Specifically, when two arrays were used for image formation, all of their signals were processed together in a single beamforming operation.

Unless otherwise stated, PAM images were formed within a plane dimensioned 16 mm (y) $\times$ 20 mm (z), using nonuniform pixel spacing for computational efficiency. The spacing in a square region surrounding the sources was set to be no larger than one-eighth of the smallest tube or source pair separation being examined. Pixel density in the peripheral areas was increased by a factor of two out to ± 4 mm and a factor of four to fill the remainder of the image. The innate resolution of the arrays minimized the likelihood of missed map features in the coarser-imaged regions (maximum pixel size was 0.5 mm), and this was verified with spot checks under several source position scenarios.

D. Dual-Array Analytical Model

Apart from the simulations and experiments described in Sections II-A–II-C, an analytical model of the dual-array configuration may be valuable for rapid investigation of the point spread function (PSF) as a function of spread angle and individual-array characteristics. Of primary utility is the quantification of FWHM beam widths, which should only require calculation of field characteristics in a limited region of space where the array beams overlap. In this scenario, simpler approximations can be written in terms of main lobe beam projections on the imaging axes.

Using the geometry of Fig. 1(b), the lateral beam pattern B_y can be approximated as the sum of two overlapping sinc functions

$$B_y = |0.5 \cdot [\text{sinc}(s_1 k y) + \text{sinc}(s_1 k y \cos \theta) \cdot \exp(-i k y \sin \theta)]|^2 \quad (3)$$

where $k = (2\pi f/c)$, $\text{sinc} = (\sin(\arg)/\arg)$, $F = (z_{\text{focus}}/L_{\text{array}})$, z_{focus} is the depth to which the array is focused, L_{array} is the array aperture, and $s_1 = (0.154/F)$ is a scale factor of ky that approximates the lateral beam pattern of the steered array.

The first term in (3) is the fixed lateral beam of the array at 0° , with the only phase associated with the lobe-dependent polarity of the sinc function. Additional phase due to nearfield steering is expected to be small and is neglected here. The second term is the estimated width and phase of the rotated array projected on to the y -axis. Within the sinc function, scaling by cosine approximates the contraction of the argument and lengthening of the beam pattern as θ approaches 90° . At and close to this angle, the second term of (3) overestimates the length of the second array's contribution to the beam pattern, but this is not relevant for FWHM calculation—only the beam crossing region (on the order of the main lobe width of the array at 0°) is at issue.

Along the z -axis, the construction of the axial pattern B_z is similar

$$B_z = \left| 0.5 \cdot \left[\text{sinc}(s_2 k z) \cdot \exp(-i k z) \cdot \frac{z}{z_f} + \text{sinc}(s_1 k z \sin \theta) \cdot \exp(-i k z \cos \theta) \right] \right|^2 \quad (4)$$

where $s_2 = (0.019/F^2)$ for the axial beam pattern of a single array, and the z/z_f term approximates the axial shift and asymmetry in the spherical-spreading corrected PAM patterns. Here, the array at 0° contributes a long main lobe along the z -axis, while the rotated array contributes a short projected lobe over most angles of interest.

The scale factors s_1 and s_2 were chosen to match FWHM estimates for a uniformly weighted linear array [7, eqs. (4) and (5)]

$$\text{FWHM}_{y,\text{single}} = 0.89 F \lambda \quad (5a)$$

$$\text{FWHM}_{z,\text{single}} = 6.95 F^2 \lambda \quad (5b)$$

where λ is the wavelength.

Review of these expressions shows that in addition to the superposition of sinc functions, there will be a contribution to the dual-array beam patterns from phase interference. Examples are shown in Appendix B. Expressions (3) and (4) are expected to be valid for most spread angles of interest, but will have reduced accuracy near 0° and 180° due to the simplified projection terms for the rotated array. The predicted PSF FWHM values are compared with full array simulation results in Section III-B.

The results above are applicable for the data-independent TEA beamformer. Estimation of PSFs for adaptive beamformers is considerably more complicated because of the vast parameter space to cover in signal and noise characteristics, and the choice of uncertainty constraint (ϵ). Therefore, development of analytical expressions for adaptive beamformer PSFs for single and paired arrays will be addressed in a separate article.

III. RESULTS

A. Single Tube Simulations and Experiments

PAM images calculated from measured and simulated single 0.5-mm inner diameter tube data are shown in Figs. 2 and 3, respectively. All images in Figs. 2 and 3 were produced by averaging PAM results from eight FUS pulse exposures. The black line overlay in each PAM image denotes a -3 -dB amplitude contour, and image data are displayed with an order of magnitude dynamic range. A sound speed of 1482 m/s, corresponding to the expected speed in water at 20°C [28], was used for all processing.

When using a single array at 0° (see Fig. 1) to image a single tube of sources, the large aspect ratios of the PAM images in Figs. 2 and 3 are evident. For example, the ratio of half amplitude axial (z) and lateral (y) TEA beamwidths is 14.9 ± 0.59 and 14.4 ± 0.70 for the measurement and simulation, respectively. When the two arrays are used together in the TEA beamformer, the resulting images [Figs. 2(e) and 3(e)] show a cross pattern with a local peak at the apparent beam intersection. This is expected when adding the beam patterns of two directional transducers. The z -axis patterns [Figs. 2(g) and 3(g)] show a substantial narrowing when using the array pair, with the measured axial FWHM dropping from 7.8 to 1.3 mm.

The improvements are even more pronounced when using the adaptive beamformer (RCB), as seen in the right-hand columns of Figs. 2 and 3. The measured axial FWHM falls

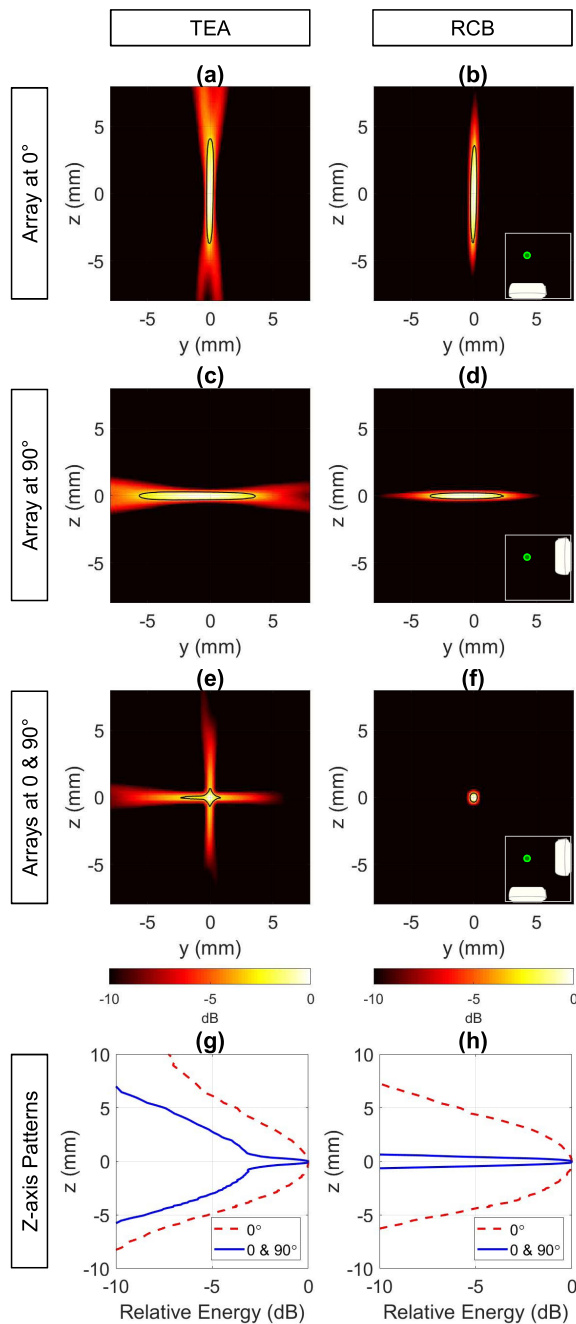


Fig. 2. Measured PAM images for a single 0.5-mm inner diameter tube with flowing SonoVue, generated with TEA (left column) and RCB (right column). (a) and (b) single array at 0°. (c) and (d) single array at 90°. (e) and (f) array pair. Array orientation schematics are shown in the lower right of (b), (d), and (f). Black lines in images (a)–(f) indicate a -3 -dB contour. (g) and (h) Normalized axial beam patterns are taken through the $y = 0$ plane.

to 1.3 and 0.7 mm with TEA and RCB, respectively, and the image areas with a response level within 50% of the peak ($A_{1/2}$) were 0.30 ± 0.03 and 0.28 ± 0.05 mm² for measurement and simulation, both of which were similar to the tube interior cross section of 0.20 mm².

The substantial improvement observed with the RCB-beamformed dual array can be explained through examination of the array element weights. Fig. 4 shows RCB weights for imaging locations at the center of the source region

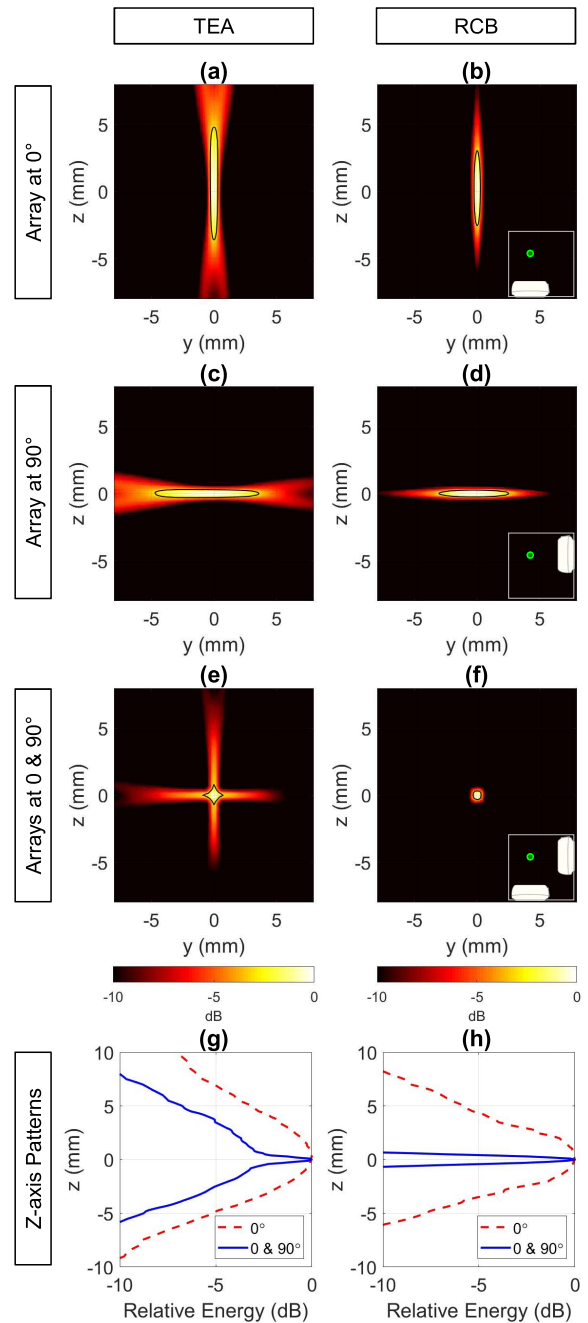


Fig. 3. Simulated PAM images for a single 0.5-mm inner diameter tube, generated with TEA (left column) and RCB (right column). (a) and (b) single array at 0°. (c) and (d) single array at 90°. (e) and (f) array pair. Array orientation schematics are shown in the lower right of (b), (d), and (f). Black lines in images (a)–(f) indicate a -3 -dB contour. (g) and (h) Normalized axial beam patterns are taken through the $y = 0$ plane.

($y = z = 0$ mm) and 1-mm off center ($y = 0$, $z = 1$ mm). When imaging the center of the source region [Fig. 4(a)], both arrays contribute equally, with an up-weighting of the end elements to sharpen the array directivity.

However, when imaging just 1 mm away from the source region [Fig. 4(b)], the adaptive beamformer essentially turns off the 0° array (which has a long PSF along the z -axis) and instead uses an amplified output from the array at 90°, taking advantage of its superior lateral resolution along

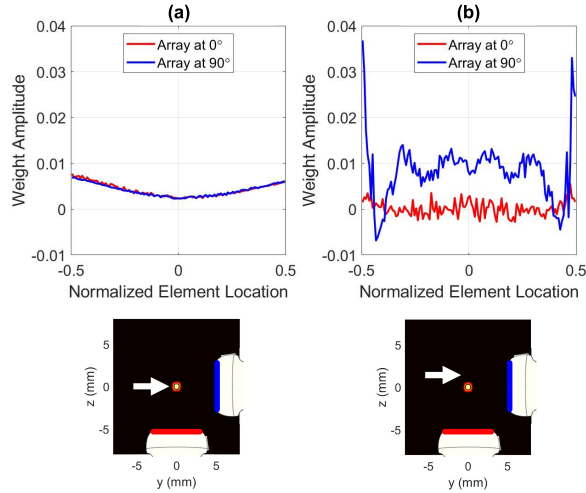


Fig. 4. RCB array element weights for dual-array simulation in Fig. 3. Element positions are normalized relative to the full array aperture. Element weights are shown for imaging pixels with (a) $z = 0$ and (b) $z = 1.0$ mm, both with $y = 0$. Diagrams below each line plot indicate array positions (not to scale) and z -axis imaging locations (white arrows) for which weights are reported.

the z -axis. This type of behavior is observed throughout the imaging plane (RCB independently determines weights at each imaging location), and it underscores the ability of RCB to enhance cavitation mapping performance using weighting schemes that are physically justifiable relative to the problem geometry.

The results in this section indicate close agreement between measurement and simulation. This suggests that the cavitation signal model well-approximates the measured cavitation data.

B. Array Separation Angle Simulations

The foregoing cavitation experiments and simulations were performed with an array separation angle of 90° , but with the assistance of the validated array simulations and the analytical model in Section II-C it is appropriate here to review the necessity of the orthogonal design specification. Fig. 5 shows lateral and axial PSF FWHM values for depths of 50 [(a) and (b)] and 76 [(c) and (d)] mm, all for a fixed single frequency of 4.0 MHz. Each subplot contains the angular dependence found from the full array simulation results for TEA and RCB, along with the analytical pair model defined in (3) and (4). The FWHM values from (5) for a single TEA-beamformed array are also included.

Looking first at the lateral results at $z = 50$ mm, the TEA pair reaches a local minimum at 90° degrees, and all values are below the single array case. This simple behavior seems to follow that of the interference pattern between the constituent beams, and is approximated by $\text{FWHM} \sim \lambda/2\sin\theta$. However, at 76 mm depth [Fig. 5(c)], the response is more complicated. Near 90° , the dual-array response is slightly wider than that of the single array. This is because the beam behaves like a sinc function with a constant offset: $B_y \sim [0.5 \cdot \text{sinc}(s_1ky) + 1]^2$. However, at wider angles, a return to the simpler angle dependence in Fig. 5(a) is seen. The transition depends on the acoustic size of the array at

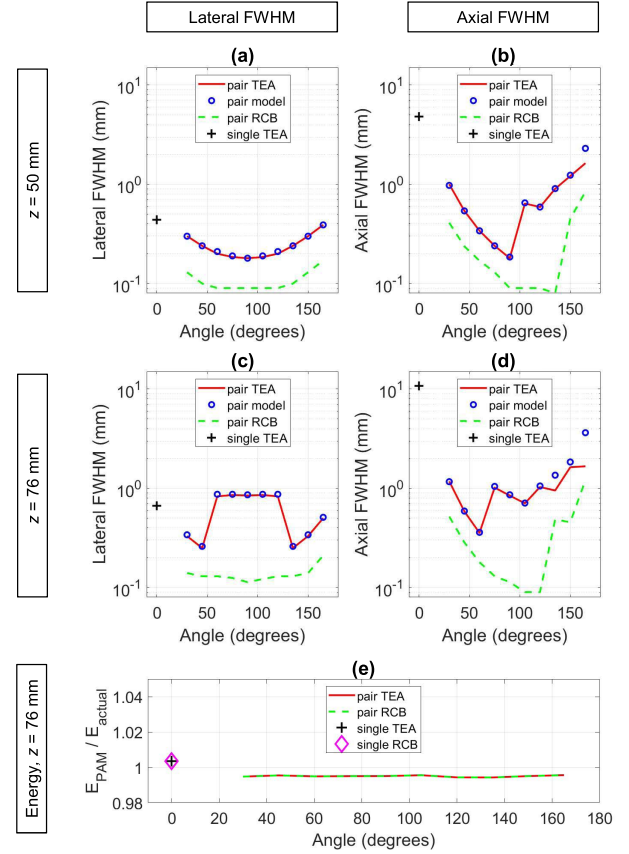


Fig. 5. (a) and (c) Lateral and (b) and (d) axial PSF FWHM as a function of array spread angle for depths of (a) and (b) 50 and (c) and (d) 76 mm, all at 4.0 MHz. Each frame shows full dual-array simulation results for TEA and RCB, along with the analytical models for TEA using an array pair (blue circles) and single array (black cross). (e) Peak map energy normalized by actual source energy.

the focal depth $\sim (k/F)$ and the beam modulation due to phase interference. Because of this interplay, there is no simple scaling law with respect to depth.

As expected, the axial FWHM in Fig. 5(b) and (d) are seen to be generally larger than the lateral FWHM, and at 90° , the axial and lateral widths are the same. In these two examples, the pair results always give a smaller FWHM than the single array. Physically this is the effect of averaging a long beam at 0° with a shorter beam from the rotated array—the combination always has a shorter FWHM than the long beam alone. Overall, the FWHM behaviors are complex, but the analytical pair model and array simulations are in good agreement. The exception is for the axial FWHM at 165° , where the beam projection expression in (4) has decreased accuracy, as expected.

RCB FWHM values in Fig. 5 are seen to be consistently lower than those obtained with TEA, with different angular dependencies. Best resolution is found at or near a spread angle of 90° and can be approximated as a quarter wavelength of the source frequency.

From Fig. 5(e), the peak map energies deviated from the actual source energy (as calculated from the point monopole emitted pressures specified in the simulations) by no more than 0.6% whether using a single array or an array pair, and

regardless of angle or beamformer. What differences exist between single and pair array results are due to the integration of background noise applied in this simulation.

Returning to the issue at the beginning of this section, the value of the orthogonal array configuration is that for any specified depth and beamformer, it provides the same FWHM in the axial and lateral dimensions. However, it is certainly possible to identify specific parameter combinations where smaller PSFs may be obtained, and the expressions in (3) and (4) may be used to quickly find these if such constrained imaging scenarios are of interest.

C. Tube Pair Experiments and Simulations

Having established the performance improvements obtainable with a pair of arrays to image a single bounded source region, imaging of a pair of source regions was investigated in order to assess resolvability and 2-D feature size. Fig. 6 shows examples of PAM results obtained from an experiment data set featuring a pair of 0.5-mm inner diameter tubes with a center-to-center spacing (Δ) of 1.5 mm. The imbalance in the brightness of the peaks near the tube centers is common with these measurements. Since the 0° array shows this imbalance, it is not due to shadowing. It could be wholly caused by a 0.35-mm lateral shift of the tube pair center relative to the centerline of the FUS beam or a 40% difference in the amount of SonoVue in each tube, although a combination of these with smaller individual errors is more likely.

In a similar manner to that seen in Figs. 2 and 3, the array pair images feature localized maxima in the expected locations, and the RCB result is superior in terms of producing “spots” that resemble the tube cross sections in the imaging plane. Specifically, the $A_{1/2}$ values were 1.37 ± 0.27 and $0.50 \pm 0.11 \text{ mm}^2$ for TEA and RCB, as compared with the actual tube pair internal area of 0.39 mm^2 . Note that the best y-direction resolution for TEA is achieved with the single array at 0° . The TEA array pair image is “filled in” between tubes as a result of the additive nature of the beamformer, such that the sidelobes from each array accumulate in the pair image.

Both beamformers report 2–5-dB lower local peak energies when using the array pair instead of a single array looking through both source regions [Fig. 6(g) and (h)]. In the particular case of the RCB array pair result, the sum of the peak values is within 7% of the single peak obtained from the single array (red dashed line), showing how that limited-aperture configuration spatially smears energy into its single axial response lobe.

Tube pair experiments were conducted for a series of decreasing tube diameters and separation distances (Δ). The results in Fig. 7 reiterate the benefit of RCB at all separation distances tested. Patterns at $z = 0$ are shown as normalized to their respective peaks so that the difference between peak and trough amplitudes is easier to visualize. Tube resolvability is quantified in Fig. 7(e) and (f) using a normalized energy ratio

$$E_{\text{ratio}} = (E_{\text{peak}} - E_{\text{trough}})/E_{\text{peak}} \quad (6)$$

where the E_{peak} value is always taken as the smaller of the two image peaks and E_{trough} is the minimum between

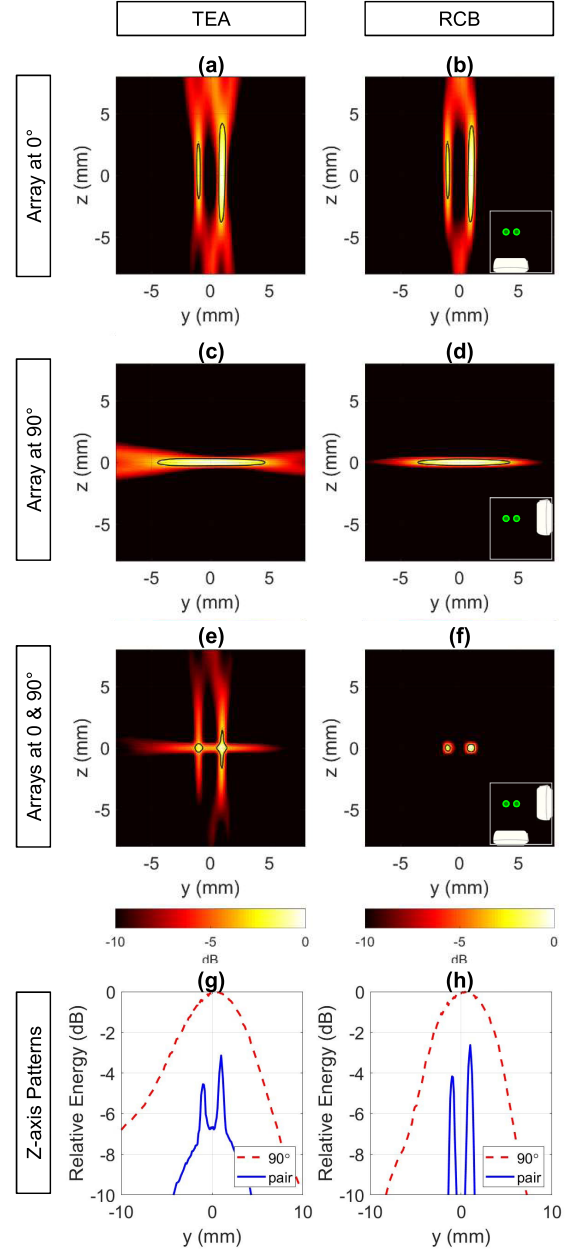


Fig. 6. Measured PAM images for a pair of 0.5-mm inner diameter tubes with center-to-center separation of 1.5 mm, generated with TEA (left column) and RCB (right column). (a) and (b) single array at 0° . (c) and (d) single array at 90° . (e) and (f) array pair. Array orientation schematics are shown in the lower right of (b), (d), and (f). Black lines in images (a)–(f) indicate a -3 -dB contour. (g) and (h) y-axis beam patterns are taken through the $z = 0$ plane and normalized by the respective peaks of the 90° result.

peaks [Fig. 7(a)]. A benchmark level of 0.263 is included, corresponding to the Rayleigh criterion [29] for resolution of two line-scatterers. By this criterion, values of $E_{\text{ratio}} < 0.263$ would be considered unresolvable. The RCB results are consistently above this threshold for all tube separations tested.

The simulation results in Fig. 7(e) and (f) show good agreement with the measurement trends except for TEA at the closest separation. This implies that overall, the tubing walls did not meaningfully modify the measured cavitation signals.

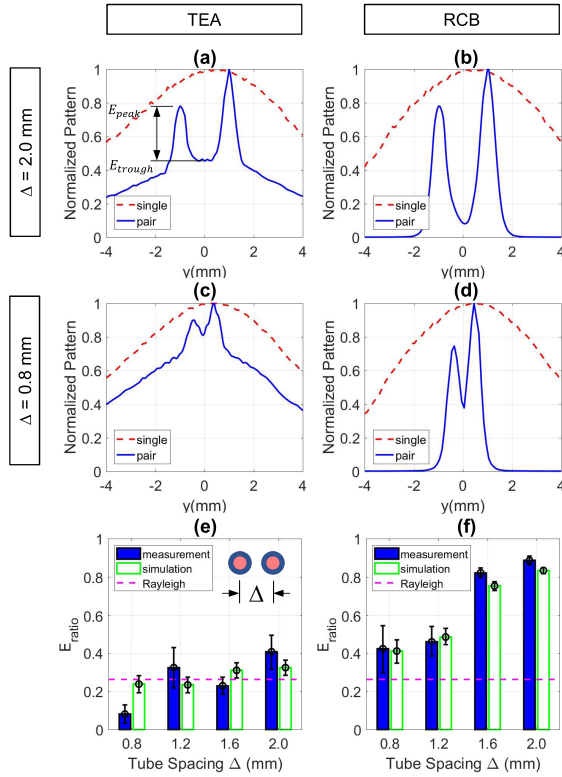


Fig. 7. Single- and dual-array measurement lateral (y) PAM patterns for (a) and (c) TEA and (b) and (d) RCB for tube center-to-center separations of $\Delta = 2.0$ (a) and (b), and $\Delta = 0.8$ mm (c) and (d). The single array was at 0° and the pair was at $0^\circ/90^\circ$. Measurement and simulation resolution are quantified in (e) and (f) through the normalized energy ratio, with the Rayleigh criterion indicated by a magenta dashed line. Values below this level suggest the sources would be considered unresolvable. Error bars indicate ± 1 standard deviation ($n = 20$).

Additional simulations (Appendix C) indicated that RCB would meet Rayleigh's criterion for $\Delta = 0.2$ mm for pairs of 0.1-mm inner diameter tubes or point scatterers. Moreover, the axial and lateral FWHMs reduce from 2.0 and 0.34 mm for a single array to 0.11 mm in both directions for the array pair.

It is important to recall that all beamformers are operating on broadband cavitation-like signals with significant energy in the 3–9-MHz band. Filtering the dual-array data with a 7.5-MHz corner frequency high pass reduces the lateral FWHM from 0.11 to 0.04 mm and allows resolution of $\Delta = 0.1$ -mm source pairs. Such a filtering operation diminishes the radiated energy estimate, so the tradeoff between resolvability and energy quantification must be evaluated for the imaging task at hand.

D. Uncertainty Simulations

The results presented thus far have presumed perfect knowledge of the most salient aspects of the imaging environment: relative position of arrays, and the absolute sound speed. The impact of global sound speed uncertainty was quantified by varying the beamformer-specified sound speed by $\pm 1\%$ – 3% relative to the assumed speed (1482 m/s in water at 20°C). The impact of relative position uncertainty was assessed by shifting the z -position and rotation angle of

the second array (nominally at 90°) by $+1$ and 2 mm, and $+5$ and 10° , respectively. Rotations were performed with respect to $y = z = 0$ in the imaging plane [Fig. 1(b)]. Finally, the cumulative effects of speed and position uncertainties were calculated. All simulations were performed using a tube pair configured with 0.4-mm inner diameter and $\Delta = 0.8$ mm.

Fig. 8 shows examples of how the above uncertainties manifest in PAM images computed with an array pair. For a $+2\%$ error in speed [meaning that the actual speed was 2% higher than assumed, Fig. 8(c) and (d)], the map is only slightly changed from the no-uncertainty case [Fig. 8(a) and (b)], with a small shifting of the individual-array patterns evident in the TEA results. This is as expected for beamformers that estimate distance using relative phase between array elements. When sound speed is presumed lower than the actual value, the algorithm shifts the source position further from each array in order to achieve the correct wavefront curvature [25]. The array PSFs therefore do not overlap at their respective peak values, resulting in image blurring and reporting of lower energy. The position error is linear in $(c_{\text{actual}}/c_{\text{presumed}})$, while the energy error is cubic, arising from the combination of mis-registration and scaling in (1).

The effects of geometric uncertainties are less subtle. Shifting the z -position of the 90° array by $+2$ mm [Fig. 8(e) and (f)] produces a mis-registration of source position of the same size but in the opposite direction. Since the array is shifted in $+z$, it will observe a peak PAM response by steering toward $-z$. By contrast, array rotation errors (without $-z$ or speed errors) of the sizes investigated are essentially negligible, whether for pairs or individual arrays (data not shown). For the case of $+2\%$ global speed error and the 90° array given a 5° rotation and $+2$ -mm z -shift, the latter is the dominant error [Fig. 8(g) and (h)].

The full set of error simulation results is compiled in Fig. 9 and indicates that all aspects of PAM are affected. Source region resolvability [Fig. 9(a) and (b)] is relatively tolerant of global sound speed uncertainty because the characteristic array beam patterns still cross without off-axis steering, and the speed-induced axial shifts (0.75–1.5 mm) are small compared to the respective individual-array on-axis beamwidths (~ 6 – 10 mm). The exception is for a speed error of 3% , which causes a large enough beam shift (~ 2.3 mm) to blur the image.

In contrast, the array position uncertainty, implemented by shifting one array relative to the other, resulted in the shifted array steering “below” the true position [Fig. 9(c) and (d)], leading to a step loss in resolvability. The impacts of combined speed and position uncertainties were examined using both positive and negative speed errors, as their corresponding beam shifts may exacerbate ($+$ %) or mitigate ($-$ %) positive z -position shifts. This is seen clearly in the $\Delta z = +2$ mm with $\Delta c = \pm 2\%$ calculations. When both errors are positive, the 90° array steers shallow in the z -direction, and the 0° array beam pattern is pushed further away in z , so that the combined z -shifts are nearly double that of the individual uncertainties. This results in a failure to resolve the source pair. Conversely, position and speed uncertainties with opposite signs have a minimal impact on source resolvability. The total area of the source regions in the imaging plane [Fig. 9(e) and (f)] is more

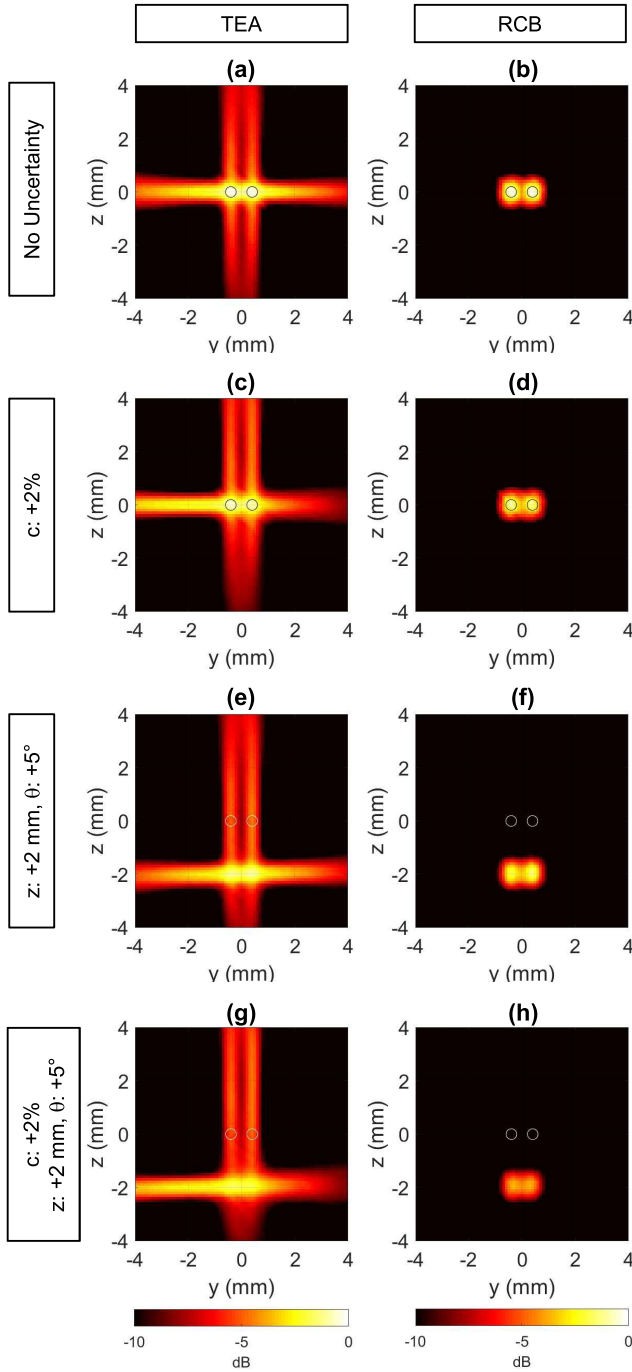


Fig. 8. Dual-array PAM image examples showing the impact of sound speed and geometry uncertainties simulated for a pair of 0.4-mm tubes, $\Delta = 0.8$ mm separation. The (a)–(d) black and (e)–(h) white circles indicate the size and position of the cavitation region boundaries in the imaging plane. Note that the y - and z -axes ranges are condensed relative to prior figures in order to more closely observe the image perturbations. (a) and (b) Baseline beamformer outputs in the absence of uncertainties. (c) and (d) Beamformer outputs using a sound speed that is 2% higher than actual. (e) and (f) Beamformer outputs after applying a +2.0 mm and +5° shifts to the array at 90°. (g) and (h) Geometry (+2.0 mm and +5°) and speed (+2%) uncertainties applied together.

sensitive to sound speed uncertainty when using TEA. The RCB results stay within 10% of the actual tube area provided that the position uncertainty is less than 2 mm. The trends in peak energy [Fig. 9(g) and (h)] generally follow those observed for resolution.

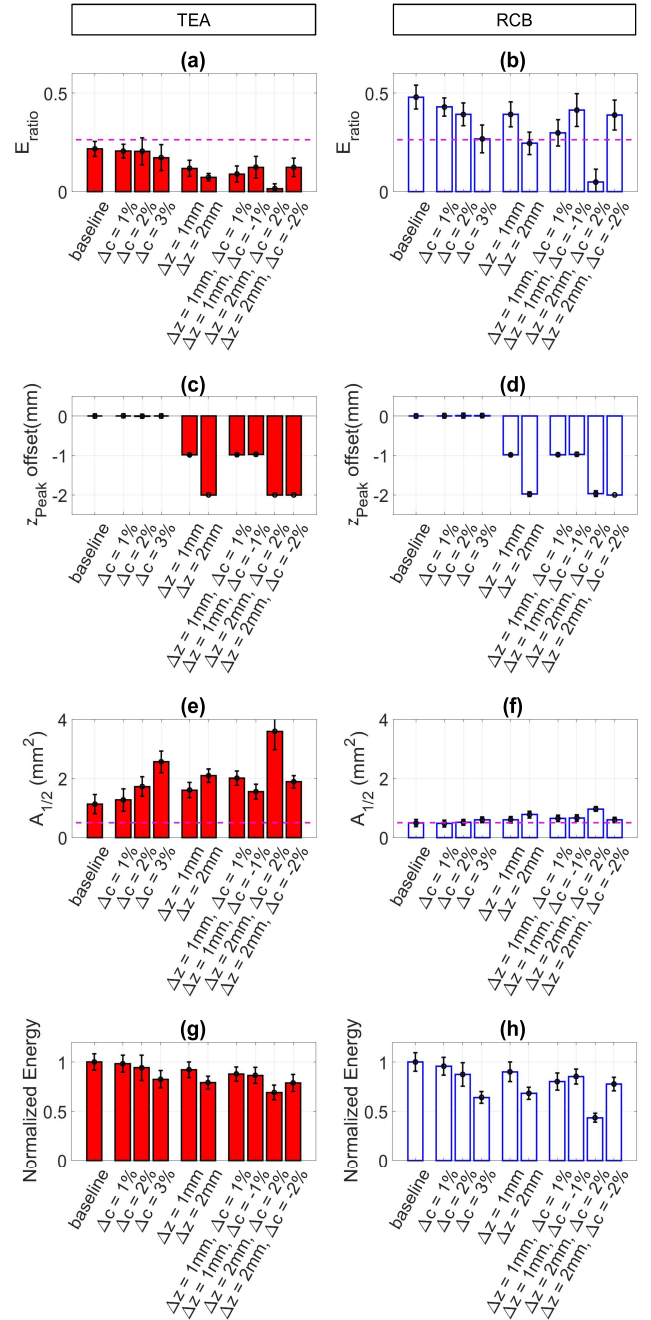


Fig. 9. Quantification of sound speed and array geometry uncertainty effects on PAM performance. Bar heights indicate means ($n = 20$, with error bars indicating ± 1 standard deviation) for simulations employing a pair of a pair of 0.4-mm tubes, $\Delta = 0.8$ mm. Values in the left column are for TEA, and in the right column for RCB. (a) and (b) Peak to trough energy ratio (refer to Fig. 7 for illustration) with reference to Rayleigh criterion (magenta dashed line) for resolvability of two line scatterers. (c) and (d) Source localization errors in the z -direction. (e) and (f) Image area with values exceeding 50% of image maximum, with reference to actual tube cross-sectional area (magenta line). (g) and (h) Map peak energy normalized to that of a single array at 0°.

Taken together, these results suggest an implementation goal of limiting position and sound speed uncertainties to the order of 1 mm and 1%, respectively.

E. Array Decimation

Adding a second array to the PAM beamforming process increases the computational burden, the degree of which varies

with the number of data channels processed and algorithm used. Simulations of simple array decimation schemes were run to assess the tradeoff in beamformer performance and channels processed, with the aim of reducing the computation time with the array pair to that of the single array. As with the uncertainty study, all decimation simulations were performed using a tube pair configured with 0.4-mm inner diameter and $\Delta = 0.8$ mm.

Element decimation schemes were based on two observations regarding peripheral elements of each array: 1) they will have the largest interelement delays (indicative of wavefront curvature) and 2) raising the peripheral element weights can sharpen directivity [e.g., Fig. 4(a)]. To preserve these features while reducing total channel count, decimation schemes were implemented such that array element density varied quadratically between the center and periphery [Fig. 10(a)]. The resulting nonuniform spacing was also expected to suppress aliasing that would otherwise arise from simpler schemes (e.g., removing every n th element).

The image calculation times are shown in Fig. 10(b), normalized to the time taken for a single array running TEA. Calculation time for TEA is seen to be approximately linear in element count, doubling for the array pair and decreasing as the elements are removed. Run times for RCB are longer and more sensitive to element count primarily as a consequence of the need to compute the eigenvalue decomposition of the covariance matrix [13]. Scaling with respect to channel count varies with computing hardware—the calculations presented here were performed on a 64-core PC running RedHat Linux.

Using 25% of the elements reduces the RCB array pair time to that of a single array with TEA, with minimal performance impact. Referring to the images in Fig. 10(c)–(h), decimation by as much as a factor of eight has a minimal impact on image clarity. A closer look at the image statistics in Fig. 10(i) and (j) reveals that TEA source resolvability actually improves slightly over the range tested, presumably because the end-weighted element selections force the algorithm to emphasize the peripheral elements and produce a sharper focus. By contrast, RCB shows a minimal sensitivity to decimation, and all results are within one standard deviation of each other for both beamformers.

The finding that such a small element count can be used to image a pair of bounded source regions is traceable to several factors. First, for the imaging distance examined here, the relative steering delays between adjacent elements are small enough that aliasing is not a concern. Moreover, for the simple scenario presented here, there is little unique phase information to be gained from adjoining elements. The broad aperture of the array pair apparently provides a sufficient diversity of wavenumbers for source identification.

The 6.25% case does exhibit elevated background and sidelobe levels, and the suitability of any of these schemes needs to be investigated in the context of the imaging problem of interest and the uncertainties that may be present. Still, it appears that it will be possible to implement the dual-array PAM without adding computation time relative to the baseline single array case.

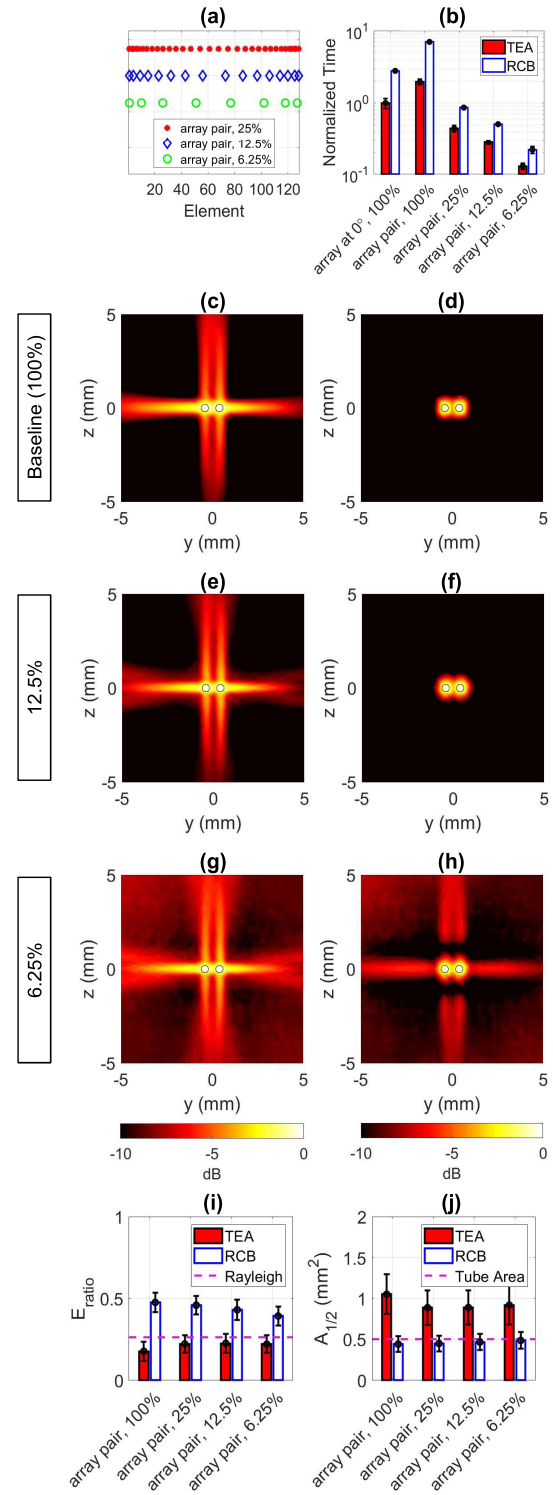


Fig. 10. Impact of array decimation on calculation time and image quality. (a) Element selection schemes. (b) Image calculation times normalized to the time for a single array running TEA. Dual-array images with (c), (e), and (g) TEA and (d), (f), and (h) RCB show minimal degradation for 8 $\times$ element decimation but 16 $\times$ decimation elevates peripheral map levels. Resolvability statistics are displayed relative to the (i) Rayleigh criterion and the (j) tube area. Error bars indicate ± 1 standard deviation ($n = 20$).

F. Ellipsoidal Volume Cavitation Simulations

Beyond fundamental investigations of distinct source pairs or regions, monitoring of cavitation emissions generated

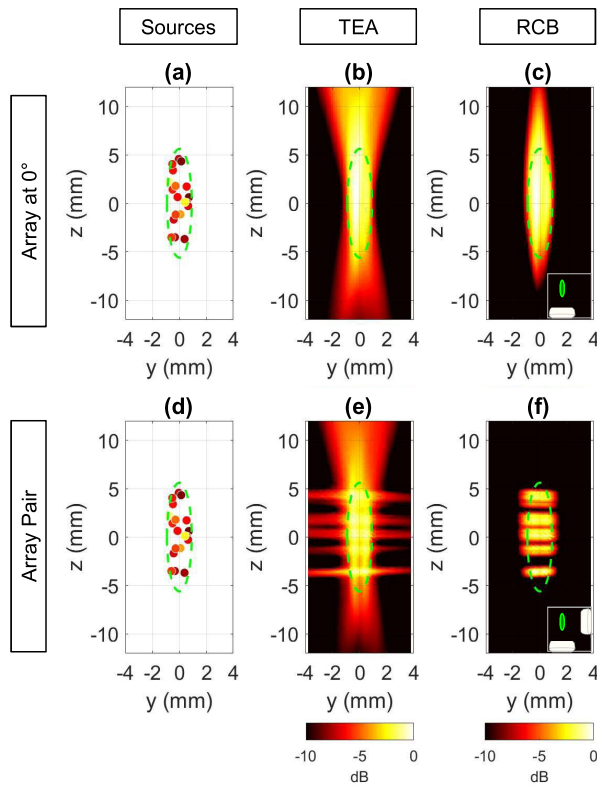


Fig. 11. PAM image examples for 20 simulated broadband sources randomly distributed within an elliptical volume (green dashed line). (a) and (d) Source layout in the imaging plane with color values indicating mean square emitted energy normalized to the peak value. (b) and (c) Images obtained using a single array at 0° . (e) and (f) Images obtained using the array pair. Array orientations are shown in the lower right of (c) and (f). In all simulations, the arrays were offset from the center of the source region by 7.6 cm, and an $8 \times 28 \text{ mm}^2$ image domain was used to visualize the single-array images.

within an expanded volume defined by the main lobe of the FUS therapy beam is relevant to treatment scenarios where cavitation agents are widely distributed (e.g., well-vascularized tissue) or not present at all, so that cavitation is driven by endogenous nuclei. This configuration was simulated with 20 sources randomly distributed within an ellipsoid corresponding to the half amplitude envelope (1.9-mm lateral, 11.2-mm axial) of the transducer used in the experiments (see Section II-A2) operating at a frequency of 1.1 MHz. Fig. 11 shows examples obtained for one random source distribution imaged with a single array and the array pair, all in the absence of array and environment uncertainties.

With a single array, the distributed sources are blended into a single response lobe whose axial FWHM exceeds the extent of the source distribution by 6.0 and 2.1 mm for TEA and RCB, respectively. The TEA array pair is able to reduce the z -extent of the image, and RCB further truncates the image in both directions, leaving highlights that correlate well with the actual source positions. Specifically, for RCB, $A_{1/2}$ reduces from 17.8 mm^2 for the single array to 5.1 mm^2 with the pair—well within the actual source bounding area of 16.7 mm^2 . While source distributions along the y -direction are not fully resolvable with the configuration and processing band employed, there is still a clear benefit in terms of map detail.

TABLE II
ELLIPSOIDAL VOLUME SIMULATION STATISTICS, $n = 20$

Quantity	Single Array (0°)	Array Pair ($0/90^\circ$)
Lateral FWHM (mm)		
TEA	1.3 ± 0.33	1.4 ± 0.36
RCB	1.2 ± 0.34	1.5 ± 0.26
Axial FWHM (mm)		
TEA	13.0 ± 1.6	6.9 ± 1.6
RCB	11.0 ± 1.8	3.5 ± 2.1
$A_{1/2} (\text{mm}^2)$		
TEA	17.0 ± 5.4	5.5 ± 2.5
RCB	12.0 ± 3.9	1.8 ± 1.2

Note: values written as $xx \pm yy$ express mean and standard deviation

The results in Fig. 11 show results for a single realization of random source positions, source strengths, and background noise. Statistics for twenty random realizations are provided in Table II. The lateral FWHM values are all less than the bounding value of 1.9 mm and are within a standard deviation of each other regardless of array geometry or beamformer. This is a consequence of the 0° array being aligned with the FUS beam envelope (Fig. 11(c), inset), so that the direction of best single array PAM resolution (y) is aligned with the narrowest cross section of the source distribution.

The advantages of the dual-array and RCB beamformer are seen prominently in the axial FWHM and $A_{1/2}$ values. For example, $A_{1/2}$ falls by nearly an order of magnitude from single-array TEA to dual-array RCB. Recalling that all cavitation emissions are generated by ideal point sources in this simulation, a smaller image area implies an improved representation of the source distribution.

IV. DISCUSSION

The measurements and simulations presented above show how the use of a pair of diagnostic arrays can substantially improve upon the 2-D image quality of PAM for a variety of source distributions. Implementation with an adaptive beamformer was especially effective, and there is evidence that the scheme could be implemented with a reduced channel count that would allow computation times to be as fast if not faster than that of a fully populated single array.

The value of fractional millimeter resolution for PAM is likely greatest for *in vitro* or small animal studies of bubble or cavitation-mediated transport of therapeutic molecules, where the relevant length scale for imaging may be as small as the intratumoral distance from the vasculature to a distal cancer cell ($\sim 100\text{--}200 \mu\text{m}$ [30]). For clinical scenarios, submillimeter resolution may not be as important as the ability to obtain a clear rendering of a distributed cavitation region (see Fig. 11). Here, the dual-array method may be valuable for image-guided therapies in limbs and thoracic organs, or for real-time feedback in cavitation memory mitigation schemes for histotripsy [31].

While several source, imaging, and environmental parameters were varied in this study, the distance from the arrays to the origin of the imaging plane was largely kept constant so that it would be easier to identify trends and sensitivities. If, for example, the arrays were placed closer to the sources of interest, the effective F -number of the imaging system

would be reduced, and the resolution should improve as allowable by the diffraction limit. Examples are shown in Fig. 5 and Appendix D.

Similarly, the background noise levels applied in the simulation study were held fixed to match typical observations from the experiments. Reducing the array element SNR is expected to adversely affect source resolvability. For example, when the mean element SNR drops from -12 to -18 dB, the simulated tube pair ($ID = 0.4$ mm, $\Delta = 0.8$ mm) becomes unresolvable for depths of 54 and 76 mm. Examples of this are also shown in Appendix D.

All measurements and simulations were performed with sources close to the point of array axial intersection ($y = z = 0$, see Fig. 1), and the imaging field of view (FoV) was typically constrained within a 16×20 mm² domain. Simulations conducted for source tube pairs displaced a lateral distance (δy) from the imaging plane origin showed negligible changes for $\delta y = 10$ mm, with gradual loss of performance out to $\delta y = 30$ mm (data not shown). At this largest distance examined, which is broader than the half-width of the arrays used and requires steering off-axis by approximately 21° , RCB images still met the Rayleigh criterion and produced $A_{1/2}$ values no greater than the actual tube area. If larger fields of view are of interest, further work would need to be done to establish sensitivities to configuration uncertainties and to assess the suitability of candidate array element decimation schemes.

The emphasis of the dual-array concept analysis to date has been on improving PAM resolution, particularly to reduce the large aspect ratios encountered with single conventional array PSFs. A potential added benefit of the increased aperture is an expansion of the FoV, as has been reported for B-mode imaging [32]. However, the value of this expansion would depend on the intended application. For *in vitro* [21] and ongoing preclinical *in vivo* work, only a small portion of the overlapping FoVs of the array pair (which is necessarily smaller than a single array FoV) has been needed. For clinical therapeutic FUS applications where it is of interest to limit potential off-target damage, evidence of cavitation would be sought throughout the full propagation path of the FUS beam, including pre- and postfocally. The extent to which a dual- or multiarray PAM configuration could enhance off-target monitoring appears to be heavily tied to the FUS source configuration, and is the subject of ongoing research.

The robustness simulations indicated sensitivity to modest uncertainty (1–2 mm) in array relative positioning. The use of alignment fixtures and procedures such as those employed for the experiments described here should provide the required positioning precision. This, in combination with environmental data (temperature or direct sound speed measurement) should suffice for minimizing uncertainty errors in simple laboratory experiments.

However, application to *in vivo* imaging, particularly clinical scenarios, will be much more challenging. Chief among the concerns is the expected difference in propagation speeds from the cavitation region(s) to the individual arrays. As an example, initial simulations performed with path speeds of the two arrays of 1505 and 1535 m/s indicated that $\Delta = 0.8$ mm

tube pairs can still be resolved with RCB for any single assumed sound speed between 1490 and 1550 m/s (data not shown), with the minimum E_{ratio} of 0.36 occurring at the highest assumed speed. PAM response characterization and compensation under conditions of uncertain differential path propagation are the subjects of ongoing work. Dual-array localization techniques established for B-mode systems [32] may be of particular value for mitigating the environmental uncertainty issue.

Beyond the challenges of propagation and position uncertainty, it is unlikely that any one array pair configuration will be well-suited for all treatment scenarios. Fortunately, the broad range of spread angles for which the 2-D resolution of the dual array is superior to the single array (see Section III-B) will allow flexibility to adjust as needed for patient morphology, source geometry, treatment target location, and physical integration into the overall therapeutic system. It is also possible that when coupled to the patient, the arrays would have different path lengths to the treatment target. Cumulative effects of the complexities mentioned here are being assessed in ongoing studies, along with methods for mitigation.

Although most of the cases examined in this study involved 3-D cavitation source distributions, all imaging results were presented in a single 2-D (lateral-depth) plane. The linear arrays used in the experiments and simulations do not allow direct examination of the third spatial dimension without physically repositioning the arrays themselves. The array elements integrate the elevation information, with spatial selectivity provided by the element height and the overlying lens. For example, measurements of elevation directivity with the L11-4v arrays indicated a half amplitude full width elevation beamwidth of 6.1 mm at 7 MHz and $z = 80$ mm. Further work will be required to enable 3-D PAM imaging with a relatively compact array configuration, with the primary goal of elevation imaging over a distance spanning the width of candidate therapy beams.

Use of arrays that were uncalibrated and contained non-functioning elements was not necessarily conducive to experimental determination of the best possible performance, but it did allow for a demonstration of robustness with respect to realistic instrumentation challenges. This was evident from the close agreement between simulations and measurements. Array calibration is essential for quantitative interpretation of PAM data [25], but this may not be necessary for applications where resolution is the primary technical requirement.

Having observed significant improvements in 2-D image quality with two hand-held arrays, investigation has begun into the PAM performance benefits of additional receiving aperture. Simulations performed with semicircular and circular arrays of the type seen in photoacoustic tomography [33] do show improvements relative to a diagnostic array pair (for example, reducing $A_{1/2}$ by more than a factor of two) for a fixed element count of 256 (Appendix E). A semicircular arrangement may indeed be well-suited for laboratory or small animal PAM studies, but further work is needed to assess the performance in candidate *in vivo* scenarios.

The numerical array simulations were independently validated with experiments and analytical models for simple source geometries. Although beyond the scope of this article, general analytical formulations describing PAM performance for arbitrary receiver geometries and selected beamformers may ultimately prove helpful for assessing other candidate configurations, and are a matter of ongoing study.

Array decimation simulations indicated the potential to accelerate PAM processing such that dual-array imaging calculations could be completed as fast as or faster than fully populated single arrays. The low sensitivity to sparsity is similar to that seen in a prior study that employed a hemispherical array [34] for transcranial imaging. Since real-time versions of TEA and RCB are already in place in our laboratory, real-time dual-array imaging is well within existing capabilities. However, the decimation study was performed with a compact source arrangement, modest array element SNR (15 dB), and no position or propagation uncertainties. Assessment of candidate PAM acceleration schemes, including alternative element spacings and optimal array sparsity under a broader range of imaging scenarios and noise conditions, is ongoing.

Finally, it is recognized that the use of two arrays is somewhat of a luxury facilitated by the availability of a flexible array controller. Implementation of a dual-array PAM scheme on a clinical scanner is conceivably straightforward, but this work has yet to be performed.

V. CONCLUSION

Dual-array PAM provides an effective means of substantially improving upon the 2-D imaging capabilities achievable with conventional ultrasound arrays. For the devices, distances, frequency ranges, and source pairs studied, PAM resolution of better than 1 mm is possible in both dimensions of the imaging plane. When cavitation data were high pass filtered at 7.5 MHz, the dual-array configuration resolved point sources separated by 100 μm . In terms of attainable 2-D resolution, dual-array PAM is in the same realm as B-mode ultrasound, CT, and MR, but with notably lower cost than the latter two modalities. No modifications to the processing algorithms are needed, and simple channel count reduction methods should enable real-time implementation. If position and propagation uncertainties can be mitigated, the proposed methods offer the potential for a step improvement in cavitation monitoring during therapeutic ultrasound procedures.

APPENDIX

A. Cavitation Signal Model

As described in Section II-B, a simplified cavitation signal model was used in this study to produce the temporal and spectral features of experimentally observed cavitation. A set of point sources was randomly distributed within a bounded volume approximating the tubes employed in the experiments. The number of point sources used in the simulations was chosen by coarsely increasing the source population until the signal trends closely matched those of the corresponding measurements. The signal from each point source was constructed from a sequence of impulses with a baseline

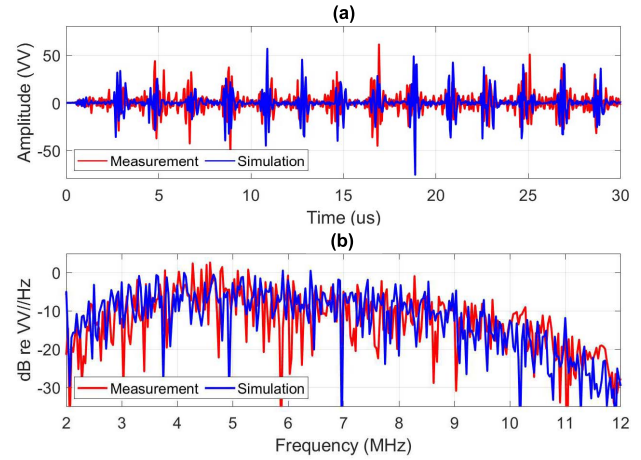


Fig. 12. Examples of array element (a) time domain and (b) frequency domain responses for measured and simulated cavitation. “VV” refers to the digitizer output values reported by the array controller.

repetition period corresponding to the FUS drive, but with each impulse randomly scaled and delayed to mimic the behavior of experimentally observed signals. Each cavitation source location was fixed over the duration of the simulated exposure. The resulting impulse train was bandpass filtered to provide a match to the typical frequency response of L11-4v array elements [25]. Array element diffraction and lens refraction terms were omitted, as they have minimal effect at the imaging distances and angles in this work [26], [34].

Fig. 12 shows an example of single element response time series (a) and spectra (b) for measured and simulated cavitation in a single 0.5-mm inner diameter tube. Simulation data were scaled to have the same time domain root-mean-square value as the measurement.

B. Array Pair Model

Fig. 13(a) and (b) shows TEA images of 4-MHz point omnidirectional source obtained for array spread angles of 90° and 150°, respectively. In the region where the beams overlap, the map intensities are highest and a clear modulation of beam intensity can also be seen. Contributions to this behavior were estimated for the lateral beam patterns (at $z = 0$) using (3) and separately examining the magnitude average and phase average of the terms in B_y .

As seen in Fig. 13(c) and (d), the red magnitude average curves show a relatively slow spatial variation, while the blue phase-interference curves show a more rapid spatial variability. Note that the phase term includes the inversion in the sinc function when going from lobe to lobe, causing the jumps seen in the blue curves. The cumulative effects are shown in Fig. 13(e) and (f), comparing full TEA array simulations to the simpler beam crossing model. These are seen to be in close agreement. Also seen here is the RCB response for $\varepsilon = 1$. It appears that the TEA and RCB beams overlay for a segment of the main lobe, then RCB quickly falls off. This behavior is being investigated separately. Axial responses in frames (g)–(j) using (4) show similar trends to the lateral responses. Overall, the interplay between the slow-varying sinc functions and the modulation produced by phase interference makes the FWHM

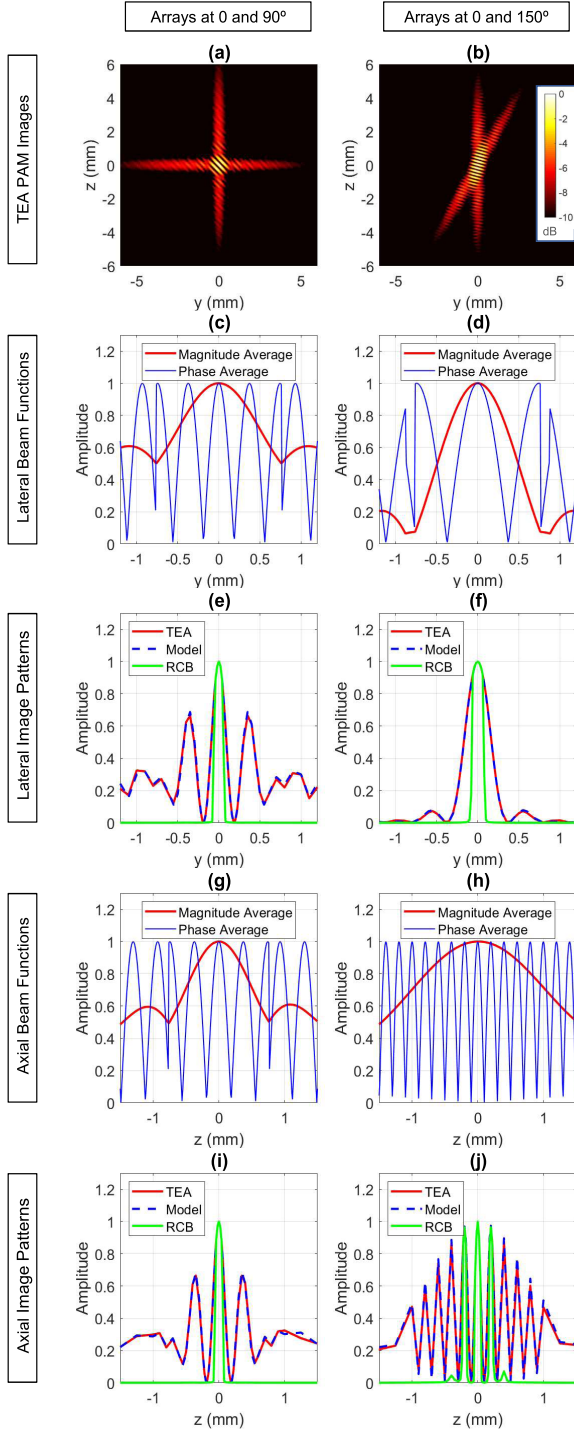


Fig. 13. TEA and beam model PAM calculations for array spread angles of 90° (left column) and 150° (right column) for a point source radiating at 4 MHz. (a) and (b) TEA images showing beam overlap and interference. (c) and (d) Lateral beam functions from (3) indicating magnitude and phase contributions to the (e) and (f) total image patterns. (g) and (h) Axial beam functions from (4) indicating magnitude and phase contributions to the (i) and (j) total image patterns.

values sensitive to angle, depth, and frequency in a way that is not simply scalable as it is for a single array.

The calculations shown here are all for a single frequency source. In the case of broadband imaging, such as is presented throughout Section III, the interference phenomena

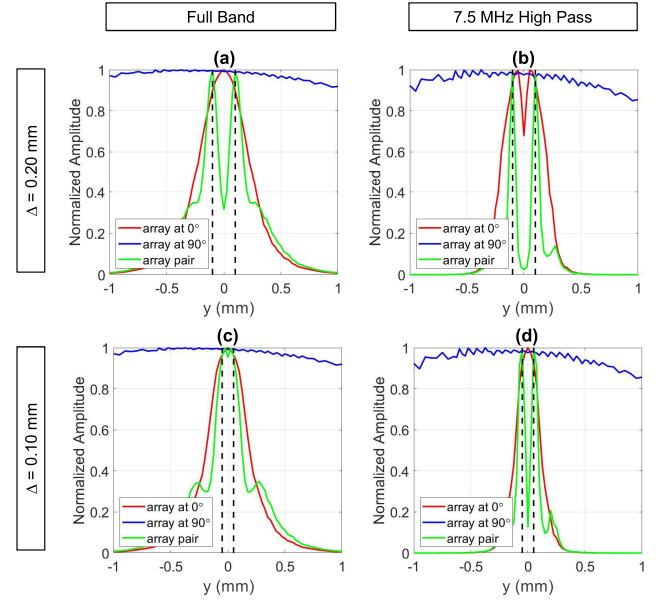


Fig. 14. Dual-array simulated lateral (y) RCB PAM patterns for point source separations indicated by black vertical dashed lines. (a) and (b) $\Delta = 0.20$. (c) and (d) $\Delta = 0.10$ mm. Results are shown for (a) and (c) full data bandwidth and (b) and (d) high-pass filtered at 7.5 MHz. Each curve was normalized by its own maximum value.

tend to integrate out, and the resulting patterns simplify to an approximately phase-less overlap of the two constituent array beams.

C. Point Source Pair Resolution

To look beyond the demonstrated resolution improvements for pairs of tube-constrained sources, the ability to resolve a pair of point sources was simulated for the case of equal amplitude radiators on the lateral (y) imaging axis. Fig. 14 shows RCB lateral patterns for source separations of $\Delta = 0.2$ and 0.1 mm, calculated using the full bandwidth of available data and after applying a fifth-order Butterworth high pass filter with a corner frequency of 7.5 MHz. The array pair enables resolution of sources at $\Delta = 0.2$ mm with full band data and $\Delta = 0.1$ mm with filtered data. Single arrays cannot separate sources under any of these tested conditions.

D. Range and SNR Effects

The effects of source-to-array distance and array element SNR on resolvability of adjoining source regions were simulated for a pair of tubes with ID = 0.4 mm, $\Delta = 0.8$ mm. Results for RCB (TEA results were similar) are shown in Fig. 15.

As expected, lateral resolution degrades linearly with increasing z -distance between array and source pair, broadening the FWHM and driving the trough to peak ratio toward unity. Independently, low SNR values fill in the trough for a fixed (z), further degrading source region resolvability. Beyond these results in a lossless fluid, deeper sources in an attenuating medium will be increasingly stripped of high-frequency information, and resolution will be coarser [25].

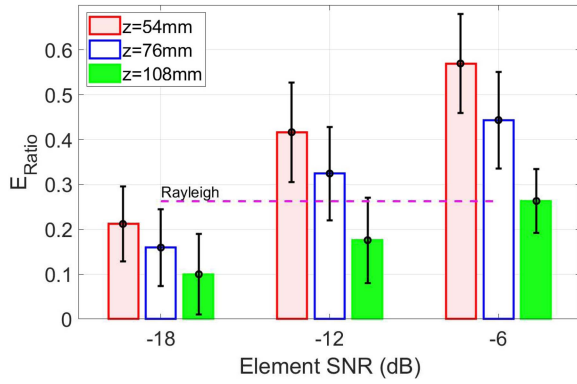


Fig. 15. Effect of range and array element SNR on source resolvability for a pair of tubes with ID = 0.4 mm, $\Delta = 0.8$ mm, imaged with RCB. Error bars represent ± 1 standard deviation ($n = 20$).

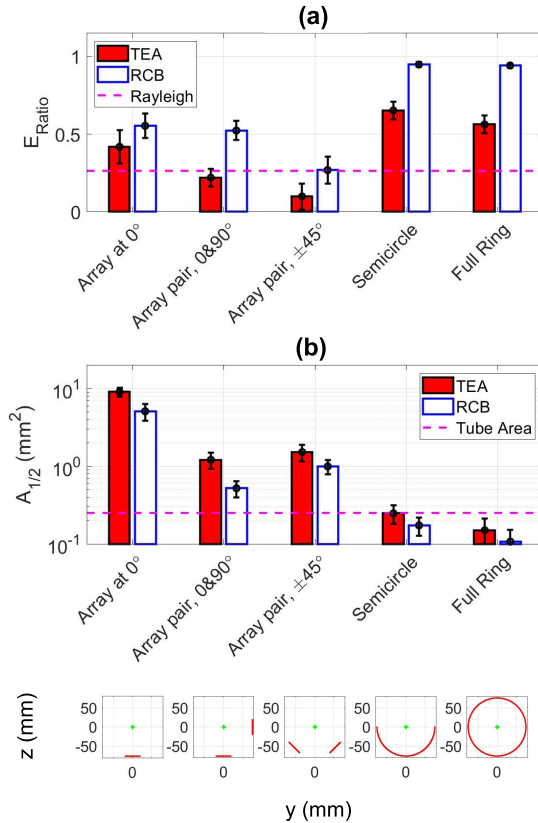


Fig. 16. Effect of array shape on source resolvability for a pair of tubes with ID = 0.4 mm and $\Delta = 0.8$ mm. With the exception of the “Array at 0°” case, all arrays had a total of 256 elements. For quantified values of (a) energy ratio and (b) image area $\geq 50\%$ of maximum, error bars represent ± 1 standard deviation for $n = 40$ random source realizations. In the array layout diagrams at the bottom of the figure, the red lines indicate receive array aperture, and the green cross indicates the center of the tube pair being imaged.

E. Alternative Array Geometries

Additional simulations were conducted to investigate the relative benefits of extended aperture arrays. Specifically, semicircular and full ring arrays with 256 elements were used to create PAM images of cavitation from the same virtual tube pair as in Fig. 15. The semicircle and ring array results in Fig. 16 indicate considerable improvement for both beamformers.

Source region resolvability approaches the maximum value (unity) with RCB, and with both beamformers, the “spot” areas are equal to or less than that of the actual tube. This is physically possible because the random distribution of sources has a mean radius that is approximately half that of the tube, and $A_{1/2}$ metric calculates the half-amplitude region of the map. So, a value for $A_{1/2}$ can conceivably be one quarter (or less) of the actual tube area. In Fig. 16(b), the RCB result for the full ring array gives an $A_{1/2}$ to A_{actual} ratio of 0.43.

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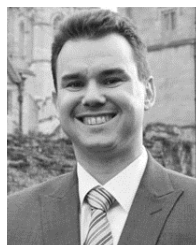
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